

switching_observer — implemented equations (H1–H5)

The Switching observer of Laquittaine & Gardner (2018) — the model that won their comparison. Labelled H1–H5 to parallel hb_integration's M1–M5, so the two documents line up block-for-block. Everything lives on the direction grid $\theta \in \{1, \dots, 360\}^\circ$, prior mode $\mu = 225^\circ$. Building block — the (normalised) von Mises pmf:

$$V(\theta; \mu, \kappa) = \exp(\kappa \cdot \cos(\theta - \mu)) / \sum_{\theta'} \exp(\kappa \cdot \cos(\theta' - \mu))$$

The one-paragraph idea: the Basic Bayesian observer MULTIPLIES likelihood and prior into one posterior and reports its mode → unimodal estimates. Subjects were BIMODAL (one peak at the stimulus, one at the prior). The Switching observer reproduces this by NOT combining them: on each trial it commits to EITHER the sensory evidence OR the prior mean, choosing between them in proportion to reliability. Because the choice is stochastic across trials, the estimate distribution has two peaks.

H1 — Evidence read-out (prior switched OFF, k_prior = 0)

Girshick MAP push-forward with a flat prior. With no prior pull, the MAP of each measurement is the measurement itself, so this is the sensory percept distribution — a bump at the stimulus whose width shrinks as coherence (k_like) grows:

$$p_e(\text{est} \mid \theta_{\text{true}}, c) = \text{GirshickMAP}(k_{\text{like}} = k_{\text{like}}(c), k_{\text{prior}} = 0)$$

Source: girshick_map_lookup(..., k_prior=0) → column at θ_{true} .

H2 — Prior read-out (likelihood switched OFF, k_like = 0)

Girshick MAP with a flat likelihood. Every measurement now maps to the prior mode, giving a spike at 225°:

$$p_{\text{prior}}(\text{est}) = \text{GirshickMAP}(k_{\text{like}} = 0, k_{\text{prior}} = k_{\text{prior}}(\text{prior})) \\ \approx \text{delta}(\text{est} - 225)$$

Source: girshick_map_lookup(k_like=0, ...) → column at θ_{true} .

H3 — Switching weights (paper Eq. 6) + lapse

Reliability-proportional competition between the two representations. With sensory reliability $k_e = k_{\text{like}}(c)$ and prior reliability $k_p = k_{\text{prior}}(\text{prior})$:

$$w_{\text{prior}} = k_p / (k_p + k_e) \\ w_{\text{evidence}} = k_e / (k_p + k_e)$$

Then the two competition weights and the FIXED lapse p_random are renormalised so the three mixture probabilities sum to exactly 1:

$$\text{total} = w_{\text{prior}} + w_{\text{evidence}} + p_{\text{random}} \\ P_{\text{prior}} = 1 - (p_{\text{random}} + w_{\text{evidence}}) / \text{total} \\ P_{\text{evidence}} = 1 - (p_{\text{random}} + w_{\text{prior}}) / \text{total} \\ P_{\text{random}} = 1 - (w_{\text{evidence}} + w_{\text{prior}}) / \text{total}$$

High coherence (large k_e) → evidence usually wins → veridical estimates. Low coherence → prior wins more → estimates biased toward 225°.

Source: _switching_weights().

H4 — Percept mixture (commit to one representation, per trial)

The percept distribution is the reliability-weighted MIXTURE of the two read-outs plus the lapse floor — NOT a product:

$$P(\text{percept}) = P_{\text{evidence}} \cdot p_e + P_{\text{prior}} \cdot p_{\text{prior}} + P_{\text{random}} \cdot (1/360)$$

(Guard: if $k_e = k_p = 0$ the observer is at chance → uniform.) The stochastic per-trial commitment to evidence-or-prior is what makes this a MIXTURE, and the two components are what make the estimate distribution bimodal. Source: estimate_distribution() step 4.

H5 — Motor noise (paper Eq. 7)

Circularly convolve the percept with von Mises motor noise centred on 0:

$$p(\text{est} \mid \theta_{\text{true}}, c, \text{prior}) = P(\text{percept}) \blacksquare V(\cdot; 0, k_{\text{motor}})$$

then renormalise to a probability vector. Source: _apply_motor_noise().

Fitted parameters (9) — the SAME set as the Basic Bayesian observer

k_like(0.06), k_like(0.12), k_like(0.24)	sensory reliability, per coherence
k_prior(80), k_prior(40), k_prior(20), k_prior(10)	prior reliability, per block width
k_motor	motor-noise concentration
p_random	lapse rate

(N_PARAMS = 9.) Because the parameter set is identical to the Basic Bayesian (integration) observer, switching buys the bimodality "for free" — no extra parameters relative to the model it beats.

How this contrasts with hb_integration (M1–M5)

Same two ingredients (a sensory read-out and a prior at 225°), combined by OPPOSITE rules:

- Switching (H4): MIXTURE — commit to evidence OR prior each trial, weights from the reliability ratio $k_p/(k_p+k_e)$. Bimodality is IMPOSED by the stochastic commitment.
- hb_integration (M4): INTEGRATION — one posterior per κ (likelihood $\times$ mixed prior), MAP read-out, averaged over the learned belief $b_t(\kappa)$. Bimodality EMERGES from the uniform floor of the mixed prior, with no switch.

Two structural differences beyond the combination rule:

- Learned latent. Switching's k_p is a FIXED fitted constant per block width (no trial-by-trial dynamics). hb_integration LEARNS the prior concentration online as a belief $b_t(\kappa)$ updated from feedback (its M5) — the promotion of a fixed parameter into an evolving latent is the whole point of the hb_integration extension.
- Weight origin. Switching's reliance on the prior is a DERIVED ratio $k_p/(k_p+k_e)$ (Eq. 6), not a free parameter. hb_integration's mixture weight α IS a fitted parameter, held fixed across blocks.